De Mathematics Competitions

3rd Annual

# DIME III

De Invitational Mathematics Examination III

Friday, September 25, 2026



## INSTRUCTIONS

1. DO NOT TURN THE PAGE UNTIL YOUR COMPETITION MANAGER TELLS YOU TO BEGIN.
2. This is a 15-question competition. All answers are integers ranging from 000 to 999, inclusive. Make sure each of your answers is 3 digits, even if the first digit is zero.
3. Mark your answer to each problem on the answer sheet by completely filling the circle with a #2 pencil. Check blackened answers for accuracy and erase errors completely. Only answers that are properly marked on the answer sheet and appear sufficiently dark will be scored.
4. SCORING: You will receive 1 point for each correct answer, 0 points for each problem left unanswered, and 0 points for each incorrect answer.
5. Only blank scratch paper, rulers, compasses, and erasers are allowed as aids. Prohibited materials include calculators, smartwatches, phones, computing devices, protractors, and graph paper.
6. Figures are not necessarily drawn to scale.
7. You will have 3 hours to complete the competition once your competition manager tells you to begin, though all work must be submitted by Friday, January 29, by 11:59 pm ET regardless of starting time.
8. You may take the DIME as many times as you want. Taking both the DIME I and II is highly encouraged.

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The problems and solutions for this DIME were prepared
by the DMC Committee under the direction of:

Taiki Aiba and Ryan Shin

The DMC Committee reserves the right to disqualify scores if it determines that the rules were not followed. Participants are subject to the DMC Policies on Competition Integrity, which can be found on our website.

The publication, reproduction or communication of the problems or solutions of this competition during the period when students are eligible to participate seriously jeopardizes the integrity of the results. Dissemination via phone, email, or digital media of any type during this period is a violation of the competition rules.

**Problem 1**:
Charles has some marbles. Their colors are either red, green, or blue. The total number of red and green marbles is 38% more than that of blue marbles, and the total number of green and blue marbles is 150% more than that of red marbles. Suppose that the total number of blue and red marbles combined is more than that of green marbles by $n\%$. Find $n$.

**Problem 2**:
Six red blocks and six blue blocks are each numbered from 1 to 6. Find the number of ways to form six unordered pairs of one red block and one blue block such that the product of the two numbers on the blocks in every pair is divisible by at least one of 2 and 3.

**Problem 3**:
Let $a$ be the positive real number that satisfies

$$\log_2 5 \cdot \log_3 6 \cdot \log_4 7 \cdot \log_5 8 = \log_2 a + \log_3 a.$$

Find $a^2$.

**Problem 4**:
There are 7 balls in a jar, numbered from 1 to 7. First, Daniel takes $a$ balls from the jar at once, where $a$ is an integer between 1 and 6, inclusive. Next, Michelle takes $b$ of the remaining balls from the jar at once, where $b$ is an integer between 1 and the number of balls left, inclusive. Finally, Katherine takes all of the remaining balls from the jar at once, if any are left. Find the remainder when the number of possible ways for this to occur is divided by 1000, given that it matters who gets which ball.

**Problem 5**:
In convex pentagon $ABCDE$ with $AE = 3$ and $DE = 26$, let $M$ be the midpoint of side $\overline{DE}$. Suppose that line $BM$ splits $ABCDE$ into two isosceles trapezoids $ABME$ and $CDMB$ such that each one has exactly three sides of equal length. Find the area of $ABCDE$.

**Problem 6**:
Richard thinks of a positive integer $n$ and writes the decimal representations of $n!$ and $(n+1)!$ on a board. He then erases any zero digits to the right of the last nonzero digit of each number, resulting in two numbers $a$ and $b$, with one of $a$ and $b$ being 4 times the other. Find the sum of all possible values of $n$ less than 1000.

**Problem 7**:
Taiki has an infinite row of empty boxes numbered $1, 2, 3, \ldots$ and an infinite supply of marbles. Each minute, Taiki finds the least positive integer $k$ such that box $k$ is empty. Then, Taiki puts a marble into box $k$, and if $k \geq 3$, then he removes one marble from each of boxes 1 through $k-2$, inclusive. Find the least positive integer $n$ such that after $n$ minutes, both boxes 9 and 10 have at least one marble in them.

**Problem 8**:
In $\triangle ABC$ with $AB = 3$ and $AC = 6$, let $D$ be the intersection of the line that bisects $\angle BAC$ and $\overline{BC}$, and let $M$ be the midpoint of $\overline{AC}$. Let the circumcircle of $\triangle DMC$ intersect line $AD$ at $P \neq D$. Suppose that $DM = 2$ and $PC^2 = \frac{m}{n}$, where $m$ and $n$ are relatively prime positive integers. Find $m + n$.

**Problem 9**:
Let $S$ be the set of all real numbers $a$ for which there exists a positive real number $b$ such that

$$\lfloor ab^2 \rfloor = 1, \quad \lfloor a^2b^2 \rfloor = 2, \quad \text{and} \quad \lfloor a^3b^2 \rfloor = 6,$$

where $\lfloor r \rfloor$ denotes the greatest integer less than or equal to a real number $r$. As $a$ approaches the greatest lower bound $A$ of $S$, the set of all possible values of $b$ approaches a set with a single element $B$. Then $A^4 + B^4$ can be written as $\frac{m}{n}$, where $m$ and $n$ are relatively prime positive integers. Find $m + n$.

**Problem 10**:
A positive integer $n$ is called *un-two* if no ordered triple of integers $(a, b, c)$ exists such that exactly two of

$$\frac{7a + b}{n}, \ \frac{7b + c}{n}, \ \frac{7c + a}{n}$$

are integers. Find the sum of all un-two positive integers.

**Problem 11**:
Quadrilateral $ABCD$ has $AB = BC = CD = 10$, $AC + BD = 24\sqrt{2}$, $\angle A$ and $\angle D$ acute, and $\angle B$ and $\angle C$ obtuse. Suppose that the distance between the midpoints of $\overline{AC}$ and $\overline{BD}$ is 5. Find $AD^2$.

**Problem 12**:
Let $z_1, z_2, z_3, \ldots, z_{2027!}$ be the distinct roots of $x^{2027!} - 1$. Suppose that $n$ is the greatest integer such that

$$\sum_{k=1}^{2027!} \frac{2^{2026!} - 1}{z_k^{2027} + 2}$$

is divisible by $2^n$. Then $n$ can be written as $a! + b$, where $a$ and $b$ are positive integers with $a$ maximized. Find the remainder when $a + b$ is divided by 1000.

**Problem 13**:
In $\triangle ABC$ with $AB = 7$, $BC = 8$, and $CA = 9$, let $D$ be on $\overline{BC}$ with $\overline{AD} \perp \overline{BC}$. Let $D'$ be the reflection of $D$ over the perpendicular bisector of $\overline{BC}$. Let $P$ and $Q$ be on the line through $D'$ parallel to $\overline{AC}$ with $P \neq Q$ and $\angle APB$ and $\angle AQB$ right. Then $AP + AQ = \frac{m\sqrt{n}+p}{q}$, where $m$, $n$, $p$, and $q$ are positive integers, $n$ is not divisible by the square of any prime, and no prime divides all of $m$, $p$, and $q$. Find $m + n + p + q$.

**Problem 14**:
A spinner has five sectors numbered $-1.25$, $-1$, 0, 1, and 1.25, each of which is equally likely to be spun. Ryan starts by writing the number 1 on a blank piece of paper. Each minute, Ryan spins the spinner randomly and overwrites the number currently on the paper with the number multiplied by the number the spinner lands on. The expected value of the greatest number Ryan ever writes on the paper can be written as $\frac{m}{n}$, where $m$ and $n$ are relatively prime positive integers. Find $m + n$.

**Problem 15**:
For a positive integer $n$ not divisible by 211, let $f(n)$ denote the least positive integer $k$ such that $n^k - 1$ is divisible by 211. Find the least positive integer $N$ such that

$$\sum_{n=1}^{210} nf(n)$$

is $N$ more than a multiple of 211.